

logspace () Function

logspace () Function is used to create an array with evenly spaced numbers logarithmically. The sequence starts at base ** start (base to the power of start) and ends with base ** stop.

Syntax:-

```
numpy.logspace(start, stop, num=50, endpoint=True, base=10.0, dtype=None, axis=0)
```

Where,

start – It represents starting element which will become base to the power of start ($base^{start}$)

stop – It represents ending element which will become base to the power of stop ($base^{stop}$)

num – It represents number of parts the element should be divided. Default is 50. It must be non-negative.

endpoint- If True, stop is the last element. If False, stop is not included.

base - The base of the log space.

dtype – The type of output array.

Creating Array using logspace () Function

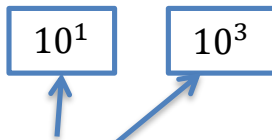
Syntax:-

```
from numpy import *
```

```
array_name = logspace(start, stop, num=50, endpoint=True, base=10.0 dtype=None)
```

Ex:-

```
a = logspace(1, 3)
```



10.0	31.62	100.0	316.2	1000.0
a[0]	a[1]	a[2]	a[3]	a[4]

```
from numpy import *
```

```
a = logspace(1, 3)
```

```
a = logspace(1, 3, 5)
```

```
a = logspace(1, 3, 5, base=12.0)
```

Accessing One-D Array Elements

```
from numpy import *
```

```
a = logspace(1, 3, 5)
```

```
print(a[0])
```

```
print(a[1])
```

```
print(a[2])
```

```
print(a[3])
```

```
print(a[4])
```

10.0	31.62	100.0	316.2	1000.0
a[0]	a[1]	a[2]	a[3]	a[4]

Accessing using for loop

```
from numpy import *  
a = logspace(1, 3, 5)
```

Without index

```
for el in a:  
    print(el)
```

With index

```
n = len(a)  
for i in range(n):  
    print(a[i])
```

Accessing using while loop

```
from numpy import *
```

```
a = logspace(1, 3, 5)
```

```
n = len(a)
```

```
i = 0
```

```
while i < n :
```

```
    print(a[i])
```

```
    i+=1
```